Dhruv Athaide

Mobile Security Engineer · Android & Flutter · Red Team · Published Researcher

Mumbai, India | +91 93206 93337 | athaidedhruv@gmail.com | linkedin.com/in/dhruvathaide | github.com/DhruvAthaide

SUMMARY

Mobile Security Engineer who has shipped **5 production Android security apps to the Google Play Store** over 2+ years. Depth in on-device threat detection, applied cryptography, end-to-end encryption (E2EE), and AOSP internals (NDK/JNI, SELinux), plus red team and mobile Vulnerability Assessment & Penetration Testing (VAPT). Co-author of a peer-reviewed IGI Global book chapter (2026).

TECHNICAL SKILLS

Languages: Kotlin, Java, Dart, C++ (NDK), TypeScript, JavaScript, Python, SQL

Android: Jetpack Compose, Flutter, Android SDK, Coroutines & Flow, Room, WorkManager, CameraX, Hilt, Retrofit/OkHttp, MVVM, Clean Architecture

Security: Applied Cryptography (AES-256-GCM, X3DH, Double Ratchet), OWASP MASVS/MASTG & Mobile Top 10, Static & Dynamic Analysis (SAST/DAST), VAPT, Threat Modeling, Secure SDLC (SSDLC), Reverse Engineering, AOSP, SELinux, JNI/NDK anti-tamper

Backend & Tooling: Node.js, Express, Firebase (FCM, Firestore, Auth), MongoDB, Azure, Docker, Git, CI/CD, Frida, Ghidra, Burp Suite

EXPERIENCE

Cyber Analyst – Red Team & Mobile Security — Mumbai, India

DeepCytes Cyber Labs UK — Aug 2025 – Present

- Own the mobile application security division end-to-end, shipping four production Android security apps to Google Play as sole or lead developer across Kotlin, Flutter, and C++ (NDK).
- Build offensive Android tooling for red team engagements mapped to the MITRE ATT&CK Mobile framework (covert telemetry, permission analysis, threat intelligence), and contribute to a hardened AOSP distribution with custom kernel modifications for high-assurance (HNI / government) builds.
- Mentor junior developers and set secure-SDLC practices (code review, release gating, native anti-tamper) across the mobile product line.

Red Team & Mobile Dev Team Lead (Internship) — Mumbai, India

DeepCytes Cyber Labs UK — Jul 2024 – Jun 2025

- Led a cross-functional team across penetration testing, threat intelligence, and vulnerability research; delivered four enterprise platforms (incl. an LMS and a SaaS product) and drove the initial architecture of the Torch remote-wipe platform.

KEY PROJECTS

Guardient – Mobile Privacy & Threat Intelligence (Play Store)

Kotlin, Android SDK, C++/NDK, Jetpack Compose, MVVM, Room

- Architected a no-root per-app firewall using Android's `VpnService` TUN interface, black-holing unauthorized app traffic at the kernel routing-table level with no throughput penalty.
- Built a deterministic permission-risk engine for toxic combinations (banking-trojan overlay, 2FA scrapers), an on-device phishing/homoglyph URL analyzer, and a quantized on-device LLM assistant (MediaPipe) – fully offline.

NoSurveil – BLE Anti-Stalking & Tracker Detection (Play Store)

Kotlin, C++/NDK, Room + SQLCipher, Hilt, BLE/GATT

- Engineered a counter-surveillance pipeline classifying 23 device types across BLE, Wi-Fi, and ASTM F3411 drone Remote-ID, with custom Apple Continuity and Google Fast Pair parsers.
- Designed a following-detection algorithm (trimmed-mean bucketing, Haversine vectors, RSSI fade thresholds) separating shadowing from stationary beacons, with RPA re-linking across rotating MACs.

LockGuard+ – Intruder Detection & Forensic Evidence (Play Store)

Kotlin, CameraX, C++/NDK, AES-256, WorkManager, Play Billing

- Built a headless CameraX capture pipeline via a synthetic `LifecycleOwner`, recording intruder photo/video on failed unlocks while the device is locked, Android 7.1–16.
- Hardened against Frida/Xposed hooks and APK repackaging via a `BiometricPrompt.CryptoObject` probe and a native C++ layer with XOR-fragmented secrets and runtime package attestation; shipped with Play Billing.

Torch – Enterprise Remote Data Protection & Device Management (Site)

Kotlin, Jetpack Compose, C++/NDK, Node.js/Express, React, Firebase, MongoDB

- Engineered a NIST SP 800-88 cryptographic erasure engine rotating across 5 ciphers (AES-CTR/CBC/CFB/OFB, ChaCha20), overwriting storage in-RAM to defeat flash wear-leveling recovery, with a 100-device fleet heartbeat service and tree-path RBAC for scoped branch administration.

SecureLine – Zero-Trust E2EE Messaging & Calling (Play Store)

Kotlin, Compose, Node.js/Fastify, WebRTC, Tor, SQLCipher, MongoDB

- Implemented the Signal protocol (X3DH + Double Ratchet over Curve25519) with hardware-backed Ed25519 keys and a CONIKS-style key-transparency verifier over append-only SHA-256 chains.
- Built WebRTC mesh voice/video with sealed signaling and SAS MITM detection, and embedded an on-device C-Tor daemon hosting ephemeral v3 onion chat rooms.

EDUCATION

Indian Institute of Technology Patna — Aug 2026 – Jul 2028

M.S. in Computer Science & Data Analytics

Amity University Mumbai — Jul 2022 – Jun 2025

Bachelor of Computer Applications (BCA) — **CGPA: 9.69 / 10.0**

RESEARCH & CERTIFICATIONS

Publication: *Design of Comprehensive Mobile Device Vulnerability Checking Application* – Book Ch. 6, IGI Global, 2026. DOI · ORCID

Certifications: Google Cloud Computing Foundations (Credly) · Google Developer Student Club Web Developer